

The authors aim to remove bias in federated learning and making it more inclusive towards lower end devices by adapting the model size for slow clients based on their hardware resources. The paper introduces Hasaas, which uses insights from the Central Limit Theorem to estimate model parameters in every round. Hasaas provides robust performance in terms of test accuracy, fairness, and convergence times compared to state-of-the-art schemes. Serving small models to slow clients and large models to fast clients offers two key benefits; reducing model training time for slow clients, and achieving a better tradeoff between model performance and convergence times compared to serving a single model to all clients. Hasaas selects a sub-model based on post-activation values of neurons after every  $r$  rounds. Neurons with smaller activation values are considered less important. Hasaas improves learns the distribution of the sample mean of the model parameters and then randomly draws each model parameter from the learned distribution rather than always using the sample mean.

Hasaas tackles both system and statistical heterogeneity by combining the benefits of differential model serving and CLT-based aggregation. Hasaas dynamically updates small models based on average post-activation values in the global model, rather than serving predefined subset models. Hasaas' design has three key components; differential model serving, i.e. the clients are served models with different sizes based on their capabilities, sub-model selection using activations; i.e. Small sub-models are selected for slow clients based on post-activation values, and model generalization using insights from the Central Limit Theorem to improve robustness over non IID datasets.

DMS has several benefits over serving the same model to all clients. Due to heterogeneity of in mobile clients, training a single model over all clients, resulting in slower clients getting dropped from FL, which introduces bias. One size fits all approach for model size can either drop accuracy or cause slow clients to get dropped. In differential model serving, Hasaas serves a random sub-model and allowing slow clients to train over the sub-model for the first  $r$  rounds. Afterwards, a new sub-model based on the post-activation values of neurons is chosen, allowing exclusion of neurons with small activation values. To tackle the statistical variance of data and generalize model training, Hasaas uses insights from CLT, and FL aims to learn the average across all clients.

The authors evaluated Hasaas on multiple models, tasks, and real-world federated datasets like CIFAR-10, Federated extended MNIST (FEMNIST), FEMNIST skewed, Fashion-MNIST for CNN and Sent140 for the LSTM model for both IID and non IID datasets, over three different learning rates. They compare Hasaas performance with FedAvg, FedProx, and Single-Model Adaptive Federated Dropout. For studying the impact of heterogeneity, the authors vary the percentage of slow devices (0%, 30%, 50%, 70%, and 90%). Hasaas achieves robust performance for different levels of system heterogeneity compared to FedAvg, FedProx and AFD. Hasaas provides better generalizability over non-IID datasets and achieves faster convergence for IID datasets. In case of high system and statistical heterogeneity Hasaas provides up to 34% test accuracy improvement over other approaches. The paper empirically shows that Hasaas also offers improved fairness.

The authors perform an ablation study and show that activation-based approach consistently outperforms random selection. The authors also found that normalizing the std. dev by inverse square root of  $t$  keeps the model stable and improves accuracy. They also observe that applying CLT to FedAvg and FedProx does not provide any significant improvement. Lastly, the authors discuss some limitations and directions for future work like studying the effectiveness of customizing model sizes for each client and exploring the impact of FL on performance of slow clients.

Overall, I find this paper to be interesting, as it solves a problem that even the big tech has not yet tackled effectively. It may be interesting to explore the threat model associated with using devices which do not support or bypass the Android remote attestation mechanism. Likewise, deciding on what model to serve to which device may not be trivial when deployed at scale. The work may also be extended to support self hosted/open source FL deployments which other devices and OSs may be able to benefit from.